Research Proposals

Joseph M. Morante Jr.

September 5, 2026

1 Reward-Model Uncertainty at the Top Tail

AT A GLANCE

Q. Among a proxy reward model's top-rated answers, can the model's own *uncertainty* flag the ones that are actually wrong?

How. Score a fixed bank of answers, take the top tail (the answers a system would pick), and check whether uncertainty adds signal *beyond* the score itself.

Test. ΔAUROC > 0 inside the selected tail, with score, length, and entropy already controlled — and whether it depends on the judges being *diverse*.

Why it matters. Uncertainty-based safety filters act exactly on this selected set. If uncertainty fails here, they inherit the proxy's blind spots — and that null is itself a safety-relevant result.

Working claim. Optimization pressure against a proxy reward concentrates selection on the proxy's own mistakes. This project tests whether reward-model *uncertainty* carries information about those mistakes—conditional on selection, beyond the proxy score itself—and whether that information is a property of ensemble *diversity* rather than of ensembling per se.

Research question. Among completions selected from the top tail of a proxy reward model, do reward-model uncertainty estimates discriminate gold-verified failures from successes *beyond the proxy score itself*, and how does this depend on *ensemble diversity*?

In plain terms. The judge's favorite answers are a mix of real winners and answers that fooled the judge. Among those favorites, does the judge's confidence add anything the score didn't already say about which are fakes—and does that depend on whether the panel of judges is near-clones or genuinely different?

1.1 Background

- **Optimizing against a proxy overfits its errors.** Best-of-n and RLHF push a policy toward the *maximum* of a learned reward, and the proxy–gold gap widens exactly where selection is strongest (Gao et al., arXiv:2210.10760). Taking the argmax of an imperfect score is precisely the operation that surfaces its mistakes, so the winner set is enriched in them.
- **Reward-model uncertainty tracks reliability—but only unconditionally.** Ensemble and quantile-head uncertainty filters low-quality completions and improves BoN selection (URM/URME,

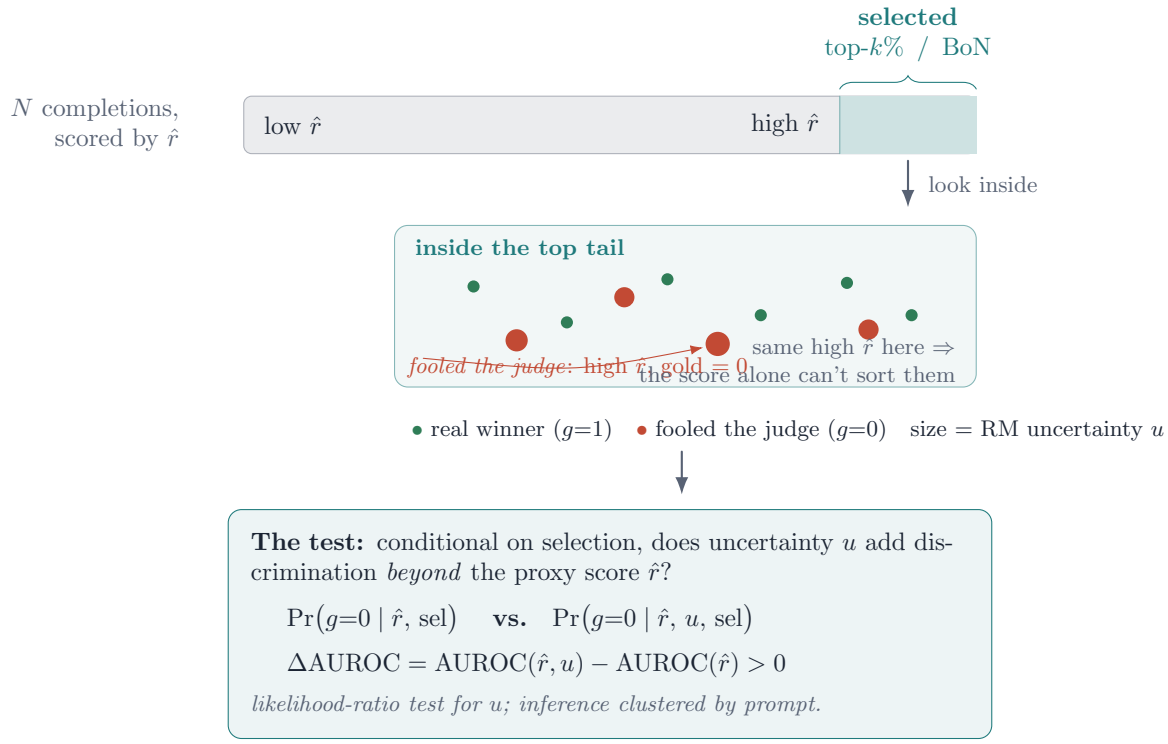


Figure 1: The top tail mixes true successes with completions that overrated the proxy. The primary estimand asks whether reward-model uncertainty u discriminates gold failures (g=0) from successes beyond the proxy score $\hat{r}$, conditional on selection ($\Delta\text{AUROC} > 0$).

arXiv:2410.00847), yet the result is reported *marginally*—over the whole distribution, at seed-only diversity (m=3), and never conditioned on the proxy score inside the tail. That marginal effect is the published anchor this proposal extends.

- **Ensembles mitigate hacking, and diversity seems to matter.** RM ensembles reduce overoptimization (Coste et al., arXiv:2310.02743); ensemble *diversity* (pretrain- vs finetune-seed) changes how much, though ensembles "mitigate but do not eliminate" hacking (Eisenstein et al., arXiv:2312.09244). Weight-averaging (WARM, arXiv:2401.12187) and uncertainty-aware training (UARM, arXiv:2606.19818; distributional / quantile RMs and their unifying lens, arXiv:2606.09073) all put uncertainty to work as *mitigation during optimization*, not as a *detector at the tail*.
- **Reward-model calibration is studied but coarse.** Uncertainty estimation for language reward models and its calibration were examined early (Gleave & Irving, arXiv:2203.07472), without a selection-conditional, tail-local, beyond-score estimand.
- **Why the tail is the regime that matters.** Uncertainty-based safeguards (filtering, abstention, selective verification) operate on the *selected* completions, where the proxy score is pinned near its ceiling (range restriction) and can no longer sort. Whether uncertainty helps *there*, and *beyond* the score, is the open and safety-relevant question.

1.2 Gap / Novelty

No published work evaluates *selection-conditional, beyond-score* detection *at the tail*, gold-verified, with ensemble *diversity manipulated and mediated*. URM/URME is the closest result and sits at seed-only diversity with an *unconditional* estimand—this study either explains its headroom or bounds its regime. The contribution is threefold and framed as tests: (i) *detection* —does RM uncertainty carry failure information beyond the proxy score, conditional on tail selection; (ii) *mechanism*—does ensemble diversity govern tail detection via error decorrelation, separating "more judges" from "different judges"; (iii) *practice*—how much detection survives cheap estimators, and what it buys in selective-verification budgets.

1.3 Hypotheses

H1 (incremental tail discrimination). Conditional on tail selection, ensemble uncertainty adds discrimination beyond proxy score ($\Delta\text{AUROC} > 0$, LR test)—and the increment shrinks as pressure grows (in n; as $k \to$ winner sets), because selection concentrates on shared blind spots.

H2 (diversity, mediated). The increment rises along D1 $\to$ D2 $\to$ D3, mediated by error decorrelation (bank- and tail-measured): a diversity step that fails to decorrelate errors fails to raise ΔAUROC.

H3 (pressure $\times$ diversity). The decay of ΔAUROC in $\log n$ is flatter for base-diverse ensembles: diversity buys robustness to optimization pressure, not just level.

H4 (cheap uncertainty). Head/dropout/Laplace/quantile-width uncertainty is tested against true ensembles on the incremental estimand at the tail; parity and gap are both headlines.

H5 (failure taxonomy + mechanism). Discrimination is heterogeneous by failure class (extremal/format/length exploits vs shared-misconception reasoning errors), and per-class error correlation across ensemble members explains it: classes where diverse members fail independently are the classes uncertainty catches.

Falsification remains informative. H1 null even at D3/D4 $\to$ "beyond-score uncertainty carries no tail information—safeguards built on it inherit the proxy's blind spots" (the safety-relevant negative). H2 flat $\to$ diversity doesn't rescue detection. H1 holds, H3 fails $\to$ uncertainty helps only under mild pressure; report the operating envelope.

1.4 Formal Problem Definition and Mathematical Formulation

Setup. For each prompt $x \in \mathcal{X}$, a fixed policy π generates a bank of N completions, scored by proxy RM $\hat{r}$. Two tail constructions, both reported: (a) *fixed-percent tails*—top $k\%$ of the bank ($k \in \{1, 5, 10\}$), comparable across n; (b) *winner sets*—the actual BoN selections (j=1 per draw) over draws of size $n \in \{16 \dots 4096\}$, the deployment-relevant set, with optimization pressure

$$\text{KL} \;\approx\; \log n - \frac{n-1}{n}.$$

Gold $g(x, y) \in \{0, 1\}$ labels tail members.

Estimand (primary). The incremental discrimination of uncertainty u over proxy score $\hat{r}$, *conditional on selection*. Fit nested logistic models of failure and compare them:

$$\underbrace{\Pr\big(g{=}0 \mid \hat{r},\ \text{sel}\big)}_{\text{base}} \qquad \text{vs.} \qquad \underbrace{\Pr\big(g{=}0 \mid \hat{r},\ u,\ \text{sel}\big)}_{\text{full}}, \qquad \Delta\text{AUROC} = \text{AUROC}(\hat{r}, u) - \text{AUROC}(\hat{r}).$$

Report ΔAUROC, partial AUC, and the likelihood-ratio test for u, with inference clustered by prompt. Unconditional tail AUROC is reported as a secondary, comparable-to-prior-work number. The selection-induced correlation between $\hat{r}$ and u inside the tail is itself reported (range-restriction diagnostic).

Uncertainty estimators u. *Primary:* ensemble dispersion (std / range / rank disagreement) over $\mathcal{E} = \{\hat{r}_1 \dots \hat{r}_m\}$. *Cheap:* multi-head, MC-dropout, last-layer/Laplace, quantile-head width. *Trivial covariates always in the model:* proxy score, length, policy token entropy.

Diversity ladder D (manipulated). D1 seed / D2 data / D3 base / D4 public-RM zoo, with diversity *measured*: pairwise error correlation on gold labels (full bank *and* tail; tail-only is selection-biased), prediction disagreement, and CKA. **Proxy quality** is a second manipulated covariate (two training-data sizes, Stage 2), so discrimination is reported conditional on how good the judge is.

1.5 Design, Statistical Plan, and Experiment Matrix

Fractional design. An anchor—D3, m=4, ensemble-std, n=1024, k=1%, verifiable gold, base proxy quality—plus one-factor-at-a-time sweeps; the full factorial is not run. **Pre-registration:** primary endpoint is ΔAUROC at the anchor cell, D3 vs D1, clustered-bootstrap 95% CI; all other contrasts are secondary under Benjamini–Hochberg FDR. **Inference:** bootstrap clustered by prompt and resampled over ensemble compositions; power target $\gtrsim$100 failures and 100 successes per primary cell for ±0.03 AUROC CI. **Confounds:** length and entropy always in the conditional model; prompt-difficulty stratification; range-restriction diagnostic; gold-noise sensitivity.

ID	Question	Design	Metrics	Tier
E1	Incremental tail discrimination (H1)	anchor ± pressure sweep (n, k, winner sets)	ΔAUROC, partial AUC, LR test vs n, k	Core
E2	Diversity ladder + mediation (H2)	D sweep $\times m$ at anchor pressure	ΔAUROC by D; error-correlation mediation	Core
E3	Pressure × diversity (H3)	D1 vs D3 × full n grid	slope of ΔAUROC in $\log n$ by D	Core
E4	Cheap uncertainty (H4)	estimator sweep at anchor	incremental ΔAUROC per estimator	Ablation
E5	Taxonomy + per-class mechanism (H5)	labeled tail failures at anchor & deepest tail	per-class ΔAUROC; per-class error corr.	Mechanism
E6	Generality	anchor + D sweep under synthetic gold	cross-design agreement	Robustness
E7	Selective verification (offline)	best cells; fixed-bank policy simulation	failures caught at b% budget; risk–coverage	Impact
E8	Proxy quality	2 quality levels × {D1, D3}	ΔAUROC × proxy accuracy interaction	Stage-2

1.6 Doubts / Risks

- **The null is publishable and safety-relevant.** If ΔAUROC ≈ 0 even at D3/D4, that bounds uncertainty-based safeguards: they inherit the proxy's blind spots. Stage 0+1 already yield this paper if the effect is flat.
- **Range restriction is the core difficulty, not a nuisance.** Selection induces $\hat{r}$–u dependence in the tail; the estimand is defined to absorb it (score always in both models) and the induced correlation is reported directly.
- **Mediation on the tail is selection-biased**—so error decorrelation is measured on the full bank as well, and both are reported.
- **E7 is offline only (scoped).** Adaptive verify-then-redraw loops create selection feedback a fixed bank cannot simulate; policy simulation is on the fixed bank in v1, with the representativeness caveat stated in limitations.

1.7 Benchmarks

- **MATH** (Hendrycks et al., arXiv:2103.03874) — primary verifiable task: exact-match gold with the proxy trained *without* gold access (weak-judge preference pairs). Stage 0 runs ∼200 prompts $\times$ N=4096.
- **HumanEval** (arXiv:2107.03374) and **MBPP** (arXiv:2108.07732) — code arm: unit-test pass/fail as gold, a second verifiable domain to check the estimand is not math-specific.
- **RewardBench** (arXiv:2403.13787) — external reference for proxy-RM quality, anchoring the manipulated proxy-quality levels (E8) to a standard scale.
- **Synthetic-gold BoN protocol** (Gao-style, arXiv:2210.10760) — open-ended generality (design b): a large gold RM labels tail members, with a gold-noise sensitivity analysis.

Metrics reported throughout: ΔAUROC and partial AUC (primary), LR test, unconditional tail AUROC (secondary), per-class ΔAUROC, error-correlation mediation, and—for E7—failures caught at a b% verification budget with risk–coverage curves.

1.8 Positioning

Work	Tail focused	Gold verified	Uncertainty as detector	Beyond score	Diversity manipulated	Taxonomy
Gleave & Irving '22 (arXiv:2203.07472)	×	×	calibration	×	observed	×
Gao et al. '22 (arXiv:2210.10760)	✓	✓ synth	×	×	×	×
Coste et al. '23 (arXiv:2310.02743)	×	✓ synth	mitigation	×	×	×
Eisenstein et al. '23 (arXiv:2312.09244)	partial	✓	mitigation	×	✓ (2 lvl)	qual.
WARM (arXiv:2401.12187)	×	✓	merging	×	implicit	×
• URM/URME (arXiv:2410.00847)	×	× (BoN qual.)	✓ uncond. filter	×	× (seed)	×
• UARM (arXiv:2606.19818)	×	×	mitigation (GRPO)	×	×	×
• Distributional RMs (arXiv:2606.09073)	×	×	theory/training	×	×	×
Lightweight-UQ / ADVPO (NeurIPS'24)	×	×	mitigation	×	×	×
This work	✓	✓ verif.+synth	✓	✓	✓ 4-lvl, mediated	✓

Open cell: selection-conditional, beyond-score detection at the tail, gold-verified, with diversity manipulated and mediated.

1.9 Value / Feasibility

Runs entirely on frozen open models: one mid-size instruct policy; Bradley–Terry proxy heads (gold withheld on verifiable tasks); the D4 zoo from public RMs; vLLM to generate the bank once and reuse it; *no RL anywhere*. The URM-style cheap baseline adds only a quantile head (same data, different head). A single node suffices. **Staged plan:** Stage 0 (vacuousness, power, confounds; gates on the proxy–gold gap and covariate saturation) → Stage 1 (pre-registered anchor + D-ladder; gate: any diversity signal promotes, flat yields a negative-result paper) → Stage 2 (fractional matrix E1–E8, both gold designs, taxonomy labeling, proxy-quality factor). Every branch produces a result.

2 LOGIC Engine: Long-Output Generation Intelligent Caching Engine

AT A GLANCE

Q. Can a reasoning model keep more of its useful "thinking" in memory by compressing *smartly* instead of uniformly?

How. Use attention entropy H and compression distortion D_{KL} to spend full precision on *logical anchors* and few bits on filler, all under a fixed *average* bit budget.

Test. At matched budget *and* GPU-hours, does adaptive precision beat static compression on final reasoning accuracy — and is it *where* the precision is spent that matters?

Why it matters. More reasoning retained per GPU-hour means better — and cheaper — reasoning models, plus a mechanistic account of which tokens must stay pristine.

Working claim. Reasoning models generate thousands of hidden "thinking" tokens, and uniform KV-cache compression destroys the few that carry the logic chain. An entropy- and distortion-aware controller that spends full-precision cache on those *logical anchors* and compresses filler aggressively—under a *matched* average bit budget—should train a more accurate reasoner per GPU-hour than any static rule.

In plain terms. Standard systems treat every part of an AI's short-term memory the same. But some tokens hold a fragile chain of reasoning and some are predictable filler. LOGIC Engine keeps a full-quality memory of the hard parts and shrinks the easy parts, fitting more useful information on the same digital scratch paper.

2.1 Background

- **Attention is structurally heterogeneous, so uniform memory rules are wasteful.** Attention profiles differ sharply across heads and layers, and adaptive, structure-aware KV compression already beats uniform schemes (**FastGen**, arXiv:2310.01801; **ARKV**, arXiv:2603.08727). A natural per-step measure of how "focused" versus "scattered" a head is at generation step t is the Shannon entropy of its attention distribution,

$$H\big(A_{l,h}^{(t)}\big) = -\sum_{i=1}^{t} P_{l,h}^{(t)}(i)\,\log_2 P_{l,h}^{(t)}(i), \tag{1}$$

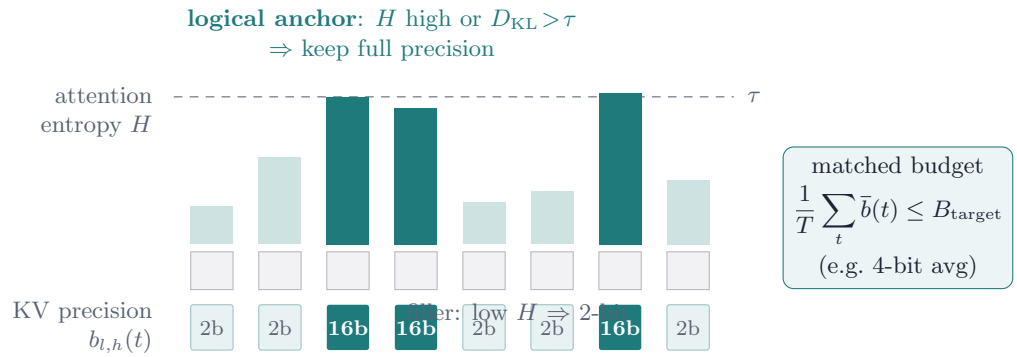


Figure 2: LOGIC Engine allocates KV-cache precision per token, head, and layer from attention entropy H and the compression-induced distortion D_{KL}: high-H / high-distortion *logical anchors* keep full precision while predictable filler is quantized hard, all under a fixed average bit budget B_{target}.

where $P_{l,h}^{(t)}(i)$ is the softmax attention on past token i. High entropy signals complex reasoning pathways whose compression risks breaking the logic chain; low entropy signals compressible, predictable text.

- **Reasoning models are different.** "Thinking" models emit thousands of hidden chain-of-thought tokens per problem, and uniform quantization silently erases vital intermediate clues (**R-KV**, arXiv:2505.24133; **LongFlow**, arXiv:2603.11504). Compression tuned for ordinary generation is not safe for a reasoning trace.
- **Logical anchors are measurable.** Recent frameworks quantify how much a head's attention *distorts* when a memory block is compressed (**KVSculpt**, arXiv:2603.27819; **RLKV**, arXiv:2510.08525), via the KL divergence between the pristine and compressed attention maps,

$$D_{\mathrm{KL}}\big(P_{l,h}^{(t)} \,\|\, \tilde{P}_{l,h}^{(t)}\big) = \sum_{i=1}^{t} P_{l,h}^{(t)}(i)\, \log_2 \frac{P_{l,h}^{(t)}(i)}{\tilde{P}_{l,h}^{(t)}(i)}. \tag{2}$$

  Tokens whose D_{KL} exceeds a safety threshold τ are flagged as *logical anchors* and shielded from compression.

2.2 Gap / Novelty

Prior KV work is either *static and uniform* (fixed-bit quantization, fixed-ratio eviction) or *structure-aware but not reasoning-budgeted* (FastGen/ARKV allocate by attention pattern; R-KV/RLKV target reasoning redundancy). None combines (i) per-token/head/layer *mixed-precision* allocation driven jointly by attention *entropy* and measured compression *distortion*, (ii) a hard *matched average-bit constraint* that makes an apples-to-apples comparison against static compression possible, and (iii) evaluation *inside the RL training loop*, where saved memory becomes more rollouts per GPU-hour and thus a better reasoner—not just a smaller inference footprint.

2.3 Hypothesis

Managing memory dynamically—shrinking conversational filler to ultra-low bit widths and reserving full precision for hard reasoning steps—lets the model retain more useful information on the same scratch paper. Over a rollout of length T, the controller enforces a strict matched average bit-rate across all layers L and heads H:

$$\frac{1}{T}\sum_{t=1}^{T}\left[\frac{1}{|L|\,|H|}\sum_{l\in L}\sum_{h\in H} b_{l,h}(t)\right] \;\le\; B_{\text{target}}, \tag{3}$$

where $b_{l,h}(t)$ is the dynamic bit precision chosen at step t and B_{target} is the controlled budget (e.g. exactly 4 bits). **H (main):** within a fixed GPU-hour budget and at the same B_{target}, entropy/KL-adaptive allocation trains a more accurate reasoner than static compression. **H (mechanism):** the gain comes from *where and when* precision is spent—anchors identified by (1)–(2)—not from a smaller footprint alone.

2.4 Experiments

1. **Adaptive vs. static at matched budget (headline).** Two identical reasoners, identical time/compute and identical *average* bit budget B_{target}: one under a rigid rule (uniform 4-bit quantization or fixed-ratio token eviction), one under LOGIC Engine. Compare final validation accuracy on hard reasoning suites, runtime, and training throughput.
2. **Does the trigger actually add value? (mechanism / ablation).** Hold total memory matched, then flip the high-precision switch *at random* for a control group; sweep the entropy and D_{KL} thresholds across layers. Test whether it is precisely *where/when* LOGIC Engine allocates pristine memory that saves the reasoning trace—or whether gains are merely a reduced footprint. The honest null (entropy/KL do not localize what matters ⇒ static is enough) is a publishable negative.

2.5 Doubts / Risks

- **Measurement overhead is the central risk.** Computing per-head H, and especially D_{KL} (which needs a pristine-vs-compressed comparison), at every step can erase the savings. A cheap proxy or periodic re-estimation, charged honestly to the budget, is required.
- **Mixed-precision KV must be implementable in the serving stack.** Per-block variable bit-width on top of PagedAttention ("virtual precision paging") needs kernel/paging support that does not negate throughput.
- **Matched-budget confound.** Average bits, GPU-hours, and wall-clock must all be held fixed, or apparent gains are footprint, not allocation—the ablation exists to detect exactly this.
- **Threshold transfer.** τ and entropy triggers may not transfer across layers or tasks; per-layer calibration and the threshold sweep address it.

2.6 Benchmarks

- **DeepSeek-R1 / OpenRLHF training benchmark** (arXiv:2501.12948; OpenRLHF, arXiv:2405.11143) — *best fit*: a clean RL loop (GRPO/PPO, cf. arXiv:2402.03300) under fixed time/space, to test

whether smarter memory yields more thoughts per second and harder problems solved.

- **vLLM inference stack** (PagedAttention, arXiv:2309.06180) — *strongest real-world proof*: building dynamic virtual-precision paging on PagedAttention shows the engine works in production and cuts serving cost.
- **Open-R1 training stack** (Hugging Face open-source reproduction) — *scalability testbed*: verify that gains found on a small model hold when scaled to multi-billion-parameter reasoners.

2.7 Value / Feasibility

Runs on frozen open reasoning models with only a small allocation controller trained on top; the heavy lifting is entropy/KL bookkeeping and quantized paging. A single node suffices for the small-model study, with Open-R1 as the scale-up path. Deliverable: an adaptive-precision KV cache that raises reasoning accuracy per GPU-hour at a fixed average bit budget, with a mechanistic account (via entropy and D_{KL}) of which tokens must stay pristine.

References

Idea 1 — reward-model uncertainty:

- Gleave & Irving. Uncertainty estimation for language reward models (2022). arXiv:2203.07472.
- Gao, Schulman & Hilton. Scaling laws for reward model overoptimization (2022). arXiv:2210.10760.
- Coste et al. Reward model ensembles help mitigate overoptimization (2023). arXiv:2310.02743.
- Eisenstein et al. Helping or herding? Reward model ensembles mitigate but do not eliminate reward hacking (2023). arXiv:2312.09244.
- Ramé et al. WARM: On the benefits of weight averaged reward models (2024). arXiv:2401.12187.
- URM / URME: uncertainty-based reward modeling and filtering (ICLR 2025). arXiv:2410.00847.
- UARM: uncertainty-aware reward modeling (2026). arXiv:2606.19818.
- Distributional / quantile reward models; unifying lens (2026). arXiv:2606.09073.
- ADVPO / lightweight uncertainty quantification for RLHF (NeurIPS 2024).
- Hendrycks et al. MATH (2021). arXiv:2103.03874. Chen et al. HumanEval (2021). arXiv:2107.03374. Austin et al. MBPP (2021). arXiv:2108.07732. Lambert et al. RewardBench (2024). arXiv:2403.13787.

Idea 2 — LOGIC Engine:

- Ge et al. Model tells you what to discard: adaptive KV cache compression (FastGen) (2023). arXiv:2310.01801.
- ARKV: Adaptive and resource-efficient KV cache management under limited memory budget (2026). arXiv:2603.08727.
- R-KV: Redundancy-aware KV cache compression for reasoning models (2025). arXiv:2505.24133.
- LongFlow: Efficient KV cache compression for reasoning models (2026). arXiv:2603.11504.
- RLKV: Which heads matter for reasoning? RL-guided KV cache compression (2025). arXiv:2510.08525.
- KVSculpt: KV cache compression as distillation — measures KL divergence between compressed-KV and full-cache outputs (2026). arXiv:2603.27819.
- DeepSeek-AI. DeepSeek-R1 (2025). arXiv:2501.12948. Hu et al. OpenRLHF (2024). arXiv:2405.11143. Shao et al. DeepSeekMath / GRPO (2024). arXiv:2402.03300.

- Kwon et al. Efficient memory management for LLM serving with PagedAttention (vLLM) (2023). arXiv:2309.06180. Hugging Face Open-R1 (open-source project).